

Batch Machine Learning

Module 4 Study Notes • Offline vs. Online Learning

1. Introduction to Production Environments

To understand the different ways Machine Learning models learn in the real world, we first need to understand the concept of **Production**. When developing an ML model, you build and train it on your local machine (Development Environment). However, for customers to actually use it, the model must be deployed to a server accessible via an IP address. This live server environment is called the **Production Environment**.

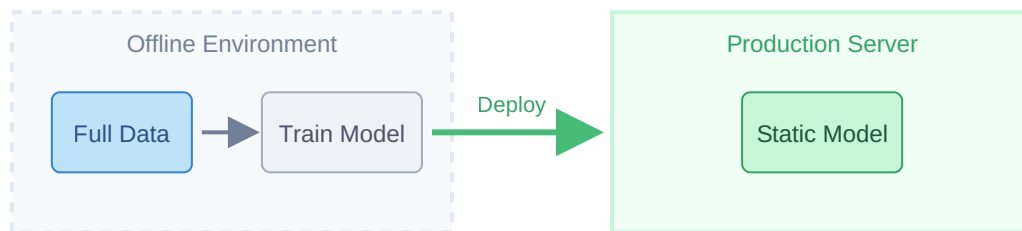
Based on how a model is trained and updated once it is destined for production, ML algorithms are divided into two categories:

- Batch Learning (Offline Learning)
- Online Learning

2. What is Batch Learning (Offline Learning)?

In **Batch Learning**, the system is incapable of learning incrementally. It must be trained using all the available data at once. Because processing large datasets requires significant time and computing resources, this training is typically done *offline* (on a local machine or dedicated computing cluster) rather than on the live production server.

- The Workflow:**
1. Collect all available data.
 2. Train the ML model offline.
 3. Test the model.
 4. Deploy the trained model to the production server.
 5. The model makes predictions on the server, but *stops learning*.



3. The Problem of Model Stagnation (Model Drift)

The biggest flaw of Batch Learning is that once the model is deployed, it is completely static. It only knows what it learned up to the moment of deployment.

However, the real world constantly evolves. For example:

- **Spam Filters:** If spammers invent a new trick tomorrow, a static spam filter won't catch it because it was only trained on yesterday's data.
- **Recommendation Systems:** If a new movie comes out, a static recommendation engine won't suggest it until it is retrained.

4. The Solution: Cyclical Retraining

To combat model stagnation in Batch Learning, engineers must set up a cyclical retraining schedule (e.g., every 24 hours, every week, or every month).

When the retraining cycle triggers, the system takes the *old data*, combines it with the *new data* that was collected since the last update, and trains a completely new model from scratch. The old model on the server is then replaced by this newly trained model.

5. Limitations & Disadvantages of Batch Learning

While Batch Learning is standard and widely used, it has three major limitations:

A. Data Volume and Computing Limitations

Because Batch Learning requires training on the *entire* historical dataset plus the new data every single time, the required computing power grows exponentially. Eventually, datasets (like social media logs)

become so massive that retraining them from scratch in one batch becomes mathematically and financially impractical.

B. Hardware and Connectivity Constraints

Batch updating requires a reliable internet connection to pull the new model from the offline environment to the production environment. If your model lives on a device with spotty or no internet (e.g., an application used by military personnel in remote locations, a satellite, or a mobile app without Wi-Fi), you cannot frequently push large model updates, rendering cyclical batch learning useless.

C. Lack of Real-Time Adaptability (Availability Delay)

If user behavior changes suddenly due to a real-world event (e.g., a major news breaking story like demonetization), users immediately want relevant recommendations. If your batch model only updates every 24 hours, it will completely miss the trend while it is happening, and will only start recommending the news the next day—when it might already be irrelevant.